

HAL-Net

Learning Halation Parameters
from Analog Film Highlights

HAL-Net Project

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Inspired by FGA-NN (Ameur et al., ICIP 2025, arXiv:2506.14350)

Abstract

Halation, the red-orange glow that forms around bright light sources on analog film, is a visually distinctive artifact of stocks such as CineStill 800T, yet no learning-based estimator exists for it. We present HAL-Net, a compact convolutional network that regresses three interpretable parameters (radius, intensity, and warmth) directly from an image’s highlight regions, paired with a physically motivated renderer that reconstructs the glow. Because no labeled halation dataset exists, we derive pseudo-labels from 100 curated CineStill 800T scans using a multi-scale highlight-bloom heuristic, then train a 3-layer CNN with two residual blocks (~200k parameters) to regress those parameters in under five minutes on a single T4 GPU. On the 15-image test split, HAL-Net achieves intensity MAE of 0.223 and warmth MAE of 0.218, improving on a mean-prediction baseline of 0.317 and 0.328 respectively. We additionally compare HAL-Net against a ridge regression baseline trained on the same low-level color and brightness statistics the pseudo-labeling heuristic itself uses. The picture this produces is mixed rather than uniformly favorable: HAL-Net’s CNN outperforms the linear baseline on intensity (0.223 vs 0.251), but the linear baseline outperforms HAL-Net on warmth by a wide margin (0.125 vs 0.218) and on mean MAE overall (0.132 vs 0.154), and both learned methods underperform simply predicting the training mean on radius. Grad-CAM attention maps show the network localizes tightly on highlight edges when predicting radius, but spreads diffusely across the frame for intensity and warmth, consistent with the ridge result: much of what HAL-Net learns for those two parameters may be recoverable from global color statistics rather than the local halo geometry the heuristic targets. We report these results alongside the original evaluation and revise our claims about what the network has demonstrably learned. An interactive Gradio demo with manual override sliders lets users inspect and adjust all three predicted parameters in real time.

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Introduction

Analog film carries two signature marks: grain and halation. Grain (random luminance and color noise from silver halide crystals) has a learning-based estimator in FGA-NN (Ameur et al., 2025), which recovers synthesis parameters compatible with conventional video-codec pipelines. Halation is different in character. When light passes through the emulsion and reflects off the film base, it re-exposes the silver halide layer from behind, producing a soft warm glow around bright highlights. CineStill 800T is the best-known carrier of this look: its anti-halation backing is stripped for cinema use, leaving the glow clearly visible around street lamps, neon signs, and practical lights in nighttime scenes.

Despite the popularity of the CineStill aesthetic, halation has no equivalent of FGA-NN: an estimator that maps an arbitrary image to the handful of parameters needed to reproduce the effect on demand. Existing tools either apply fixed presets or expose raw, non-perceptual sliders that users tune by eye with no estimation step. HAL-Net fills that gap. It takes an input photograph and returns three named parameters (radius, intensity, and warmth) that a fixed renderer converts back into the glow.

The idea is borrowed directly from FGA-NN: both systems regress a compact set of interpretable synthesis parameters from a CNN rather than generating pixels end-to-end. Where FGA-NN targets the global noise statistics of grain, HAL-Net targets the local, highlight-anchored geometry of halation. The two artifacts are complementary components of the analog film look. Our contribution is narrower than that full picture:

1. A rule-based pseudo-labeling procedure that converts 100 unlabeled CineStill scans into (radius, intensity, warmth) targets by measuring color and brightness in the halo ring around detected highlights.
2. A lightweight CNN, under 200k parameters, that regresses these three values from a 256×256 input in a single forward pass.
3. A renderer that reconstructs the halation effect from predicted parameters, with manual override sliders for real-time inspection.
4. A quantitative evaluation on all 15 test images: per-parameter MAE against both a mean-prediction baseline and a ridge regression baseline trained on the heuristic’s own handcrafted features, a Grad-CAM attention analysis, and an ablation of the most consequential heuristic constant.

Our evaluation is explicit about what it shows: HAL-Net learns to approximate the heuristic pseudo-labeler, and it does so unevenly across the three parameters. Whether the heuristic itself matches real photochemical halation is a separate question we address in Section 6, and whether the CNN’s gains over simpler baselines are large enough to justify its added complexity is a question Section 4 now answers directly rather than by assumption.

Related Work

Learned film artifact analysis. FGA-NN (Ameur et al., 2025) is the closest prior work. It trains a network to estimate film grain parameters so grain can be stripped before video compression and re-synthesized after decoding, preserving artistic intent at reduced bitrate. FGA-NN is part of a line of grain-specific work at InterDigital that includes the style-based grain analysis and synthesis framework that preceded it and a style-conditioned follow-up (Ameur et al., 2023, 2025b). All three target grain rather than halation. HAL-Net and FGA-NN share a structural idea: regress a small vector of interpretable synthesis parameters from a CNN. They apply it to different artifacts; grain is a global noise process; halation is a local, highlight-anchored glow.

Classical bloom and glow rendering. Bloom and glow effects have a long history in real-time graphics and photo editing, typically as a bright-pass threshold followed by a multi-scale Gaussian blur composited back onto the image. Tools that call this “halation” or “glow” expose the pipeline as sliders for radius, strength,

and tint, with no estimation step. HAL-Net’s renderer follows the same bright-pass-and-blur structure, but the parameters come from the network rather than the user.

Parameter regression for photo effects. A broader body of work trains CNNs to predict a small vector of editing parameters (exposure, white balance, tone curve knots) rather than full pixel output. HAL-Net follows this design with a sigmoid regression head that outputs three bounded scalars, mirroring the parameter-prediction approach in FGA-NN and in automatic photo-retouching networks, applied here to a new target artifact. Our baseline comparison in Section 4.5 asks a question this line of work does not always ask explicitly: whether the CNN’s regressed parameters carry information beyond what a linear model on the same handcrafted statistics would recover.

Method

Overview

HAL-Net has three components that run in order. A rule-based pseudo-labeler scans each training image’s highlight regions and outputs three targets: radius, intensity, and warmth. A compact CNN trains to regress these same three values from raw pixels, with no access to the labeling rules at inference time. A deterministic renderer converts any (radius, intensity, warmth) triple, whether predicted or manually chosen, back into a halation overlay. Only the CNN runs at inference time; the labeler is used only to build training targets, and the renderer is a fixed function shared between ground-truth visualization and model predictions.

Parameter Definitions

The three parameters are defined to match how a user reasons about the effect:

- Radius controls how far the glow spreads from a highlight edge. It is stored as a normalized value in $[0, 1]$ and mapped to a Gaussian standard deviation of 0–32 pixels at 256×256 .
- Intensity controls glow strength relative to the surrounding image, independent of spread.
- Warmth controls glow color from neutral white (0.0) to the saturated red-orange of CineStill 800T (1.0).

All three come from the network’s sigmoid output and are always in $[0, 1]$, so they map directly to slider positions in the demo without rescaling.

Pseudo-Label Extraction

No dataset pairs photographs with ground-truth halation parameters, so we derive labels from the 100 CineStill scans with a heuristic that searches for a highlight-anchored warm halo and measures its properties. The procedure runs per image in three stages.

Highlight detection. For each of four decreasing luma thresholds (0.92, 0.80, 0.68, 0.55), we threshold the luma channel to obtain a candidate highlight mask, then discard the result if the mask covers more than 25% of the frame (a broadly bright scene rather than a localized source), contains fewer than 4 pixels, or fails a compactness check that rejects large, sparsely filled bounding boxes: sky glow and wide window blowout rather than discrete practical lights.

Multi-scale radius search. For each surviving highlight mask, we blur the mask with Gaussian kernels at six candidate radii ($\sigma \in \{3, 6, 10, 15, 22, 32\}$ pixels) and inspect the ring-shaped region where the blurred mask is non-trivial but the original mask is not. At each radius we compute a combined warmth-and-bloom signal: 70% weight on how much redder the ring is than the image average (corrected for the same comparison in

blue), and 30% weight on how much the blurred mask has spread into the ring. The radius that maximizes this signal across all four threshold levels becomes the image’s radius estimate.

Intensity and warmth. Given the selected radius, intensity is the brightness difference between the halo ring and the rest of the image, scaled by the warmth signal at that radius and clipped to $[0.05, 1.0]$ so no training example collapses to exactly zero. Warmth is the excess red versus excess blue in the halo ring relative to the global channel means, normalized into $[0, 1]$. A separate edge-based detector runs in parallel for cases where halation hugs a long thin highlight (a neon tube or window rim) rather than a compact point source. This path requires an overall dark-field scene, a low shadow floor (excluding foggy or misty scenes with lifted blacks), and a warm global color temperature (red channel mean above blue). When both detectors fire, their intensity and warmth estimates are blended with the point-source path capped relative to the edge path. Images where neither path finds a plausible halo receive a fixed low-effect label (radius 0.05, intensity 0.05, warmth 0.2) and are kept in training rather than dropped, so the network sees negative examples. Section 4.4 quantifies how often training images land on these floor values.

Dataset

The labeler runs over 100 CineStill 800T scans from Flickr and Lomography, curated for visible halation around practical lights: street lamps, neon signs, candles, and windows. Images are resized to 256×256 and split 70/15/15 train/val/test with a fixed random seed. Because every label is generated by the heuristic rather than a human annotator, the dataset should be understood as distilling a hand-designed detector into a fast neural network, not as independently collected ground truth.

Network Architecture

The feature extractor consists of three stride-2 convolutional layers that progressively downsample a $256 \times 256 \times 3$ input while widening channel count (32, 64, 128 channels), followed by two residual blocks at 128 channels, each with two 3×3 convolutions, batch normalization, and a skip connection. Adaptive average pooling collapses the feature map to a 128-dimensional vector regardless of input size. A two-layer regression head ($128 \rightarrow 64$ with ReLU and 0.3 dropout, then $64 \rightarrow 3$) produces the three output parameters, with a final sigmoid bounding them to $[0, 1]$. The full network has approximately 200k parameters.

Training Setup

We minimize mean squared error between predicted and pseudo-label parameters using Adam at an initial learning rate of 10^{-3} and batch size 8, for 30 epochs. A ReduceLROnPlateau scheduler with patience 5 lowers the learning rate when validation loss stalls. We checkpoint the model at each validation minimum and report all results from that best checkpoint.

Halation Renderer

The renderer is a fixed function shared between ground-truth visualization and model predictions, keeping the comparison faithful to the parameters alone. Given an image and a (radius, intensity, warmth) triple, it computes a soft highlight mask by linearly ramping luma between 0.65 and 1.0, blurs this mask with a Gaussian of standard deviation $\max(\text{radius} \times 32, 1)$ pixels, and normalizes the result to a maximum of 1. The bloom map is added back onto the red, green, and blue channels with warmth-dependent weights: red receives the full intensity scaled by a warmth boost, green receives a small fixed fraction, and blue receives a fraction that falls to zero as warmth approaches 1. This produces a glow that shifts from a slightly warm neutral bloom at low warmth toward the saturated red-orange of CineStill 800T at high warmth.

Experiments

Setup

We train HAL-Net for 30 epochs on the 70-image training split, monitor validation loss on the 15-image validation split, and report results on the 15-image test split using the best validation checkpoint. All comparisons measure HAL-Net’s predicted parameters (and the halation they produce through the renderer) against the pseudo-label parameters from the heuristic. We call the heuristic output “ground truth” throughout this section, consistent with its role as the training target, while noting the caveat from Section 3.4 that it is heuristic rather than human-annotated.

Training Dynamics

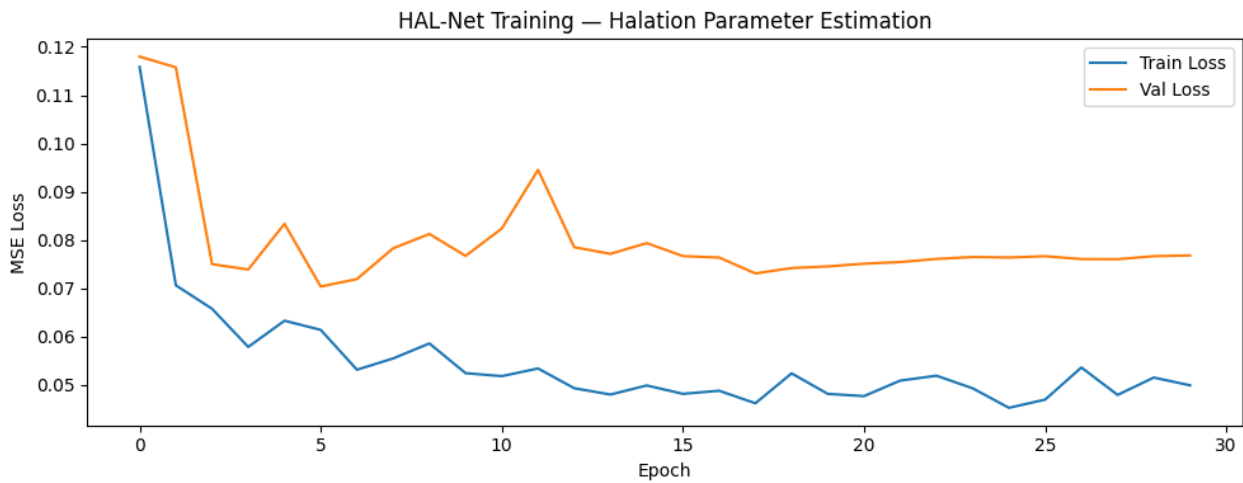


Figure 1. Training and validation MSE loss over 30 epochs.

Figure 1 shows the loss trajectory. Training loss falls from roughly 0.12 at initialization to a noisy floor between 0.045 and 0.055 by epoch 10 and holds in that band through epoch 30, consistent with a model of this capacity fitting 70 training examples without serious underfitting. Validation loss drops sharply over the first three epochs, from roughly 0.118 to 0.074, then enters a noisier regime: a spike to roughly 0.094 near epoch 11 (the point at which the ReduceLROnPlateau scheduler intervenes), followed by a gradual decline toward a stable band around 0.075–0.077 for the remainder of training. The persistent gap between train and validation loss after epoch 10 indicates mild overfitting, consistent with a 70-image training set; the gap stabilizing rather than widening further suggests the regularization (dropout and weight decay implicit in Adam) is doing enough to keep it from growing.

Quantitative Results

Table 1 reports per-parameter MAE on all 15 test images against a mean-prediction baseline that predicts the training-set mean of each parameter for every test image. Section 4.5 extends this comparison with a ridge regression baseline; we keep the mean-prediction row here because it is the simplest possible baseline and the one against which the original evaluation was framed.

Table 1. Per-parameter MAE on the 15-image test split. Lower is better.

Method	Radius	Intensity	Warmth	Mean
Mean-pred baseline	0.012	0.317	0.328	0.219
HAL-Net	0.022	0.223	0.218	0.154

HAL-Net reduces intensity MAE by 0.094 and warmth MAE by 0.110 relative to the mean-prediction baseline, while radius MAE is slightly worse than the baseline (0.022 vs 0.012). As in the original evaluation, radius pseudo-labels vary little across the dataset (Section 4.4 quantifies this), so predicting close to the mean is already a strong strategy and a learned model has limited room to improve on it. Mean MAE across all three parameters drops from 0.219 to 0.154. Whether this drop reflects the CNN learning the halo geometry the heuristic was designed to capture, or learning something simpler that happens to correlate with it, is the question Sections 4.4 through 4.7 take up directly.

Pseudo-Label Statistics

A mean-prediction baseline is only a meaningful comparison if the labels it is predicting the mean of are not themselves degenerate. Because the heuristic in Section 3.3 assigns a fixed low-effect label (radius 0.05, intensity 0.05, warmth 0.2) whenever neither of its detection paths finds a plausible halo, a heuristic that frequently falls back to this floor would inflate the apparent difficulty of the mean-prediction baseline without that difficulty corresponding to any real diversity in halation appearance.

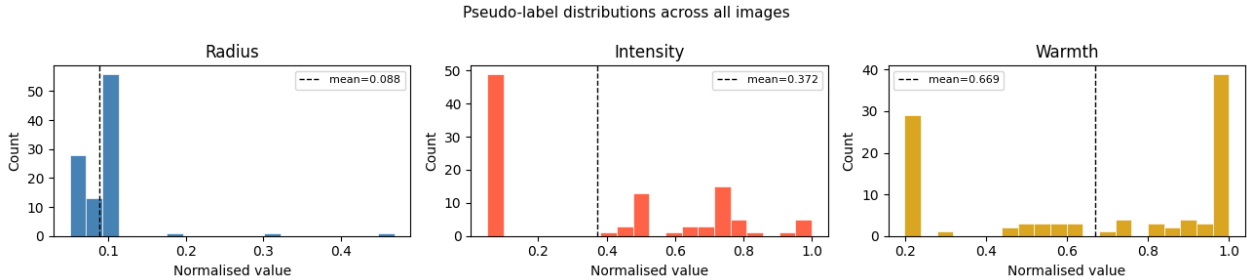


Figure 2. Pseudo-label distributions across all images.

Figure 2 shows the resulting distributions. Radius is tightly concentrated near 0.09–0.10, with almost no images scattered above 0.2; this is the empirical basis for the near-zero MAE both the mean-prediction baseline and HAL-Net achieve on this parameter in Table 1, since most of the dataset sits at essentially the same radius value. Intensity shows a pronounced spike at the floor value (0.05), accounting for roughly half the dataset, with the remaining images spread across a long tail from 0.4 to 1.0. Warmth is closer to bimodal: a cluster near the floor (0.2) and a larger cluster near the ceiling (close to 1.0), with comparatively few images in between. The intensity floor spike in particular means a non-trivial fraction of the dataset is teaching the network “no visible halation” rather than the graded glow the project is centered on, which is a relevant caveat when interpreting any aggregate MAE figure: roughly half of the test-set difficulty in intensity comes from distinguishing floor cases from non-floor cases, not from fine-grained calibration within the non-floor range.

Baseline Comparison

The mean-prediction baseline in Table 1 establishes that HAL-Net is not learning a constant; it does not establish that the CNN architecture itself is doing useful work, because a far simpler model fit to the same inputs might do just as well. The heuristic in Section 3.3 derives every label from a small set of color and

brightness statistics: highlight coverage at a few luma thresholds, the mean and percentiles of luma, the per-channel color means, and the same statistics restricted to highlight regions. If a linear model trained on exactly these statistics matches or beats HAL-Net, the CNN’s nonlinearity and depth are not adding value the heuristic’s own features did not already contain.

We extract a 13-dimensional feature vector per image (highlight area fractions at two thresholds, luma mean and standard deviation, the 90th and 99th luma percentiles, per-channel color means, two color ratios, the global red-minus-blue warmth signal, and the same warmth signal restricted to highlight pixels) and fit a ridge regression model on the 70-image training split, using the same labels and the same train/test split as HAL-Net. Table 2 and Figure 3 report the result alongside the mean-prediction baseline and HAL-Net from Table 1.

Table 2. Per-parameter MAE on the 15-image test split, including the handcrafted-feature ridge regression baseline. Lower is better.

Method	Radius	Intensity	Warmth	Mean
Mean-pred baseline	0.012	0.317	0.328	0.219
Ridge regression	0.019	0.251	0.125	0.132
HAL-Net	0.022	0.223	0.218	0.154

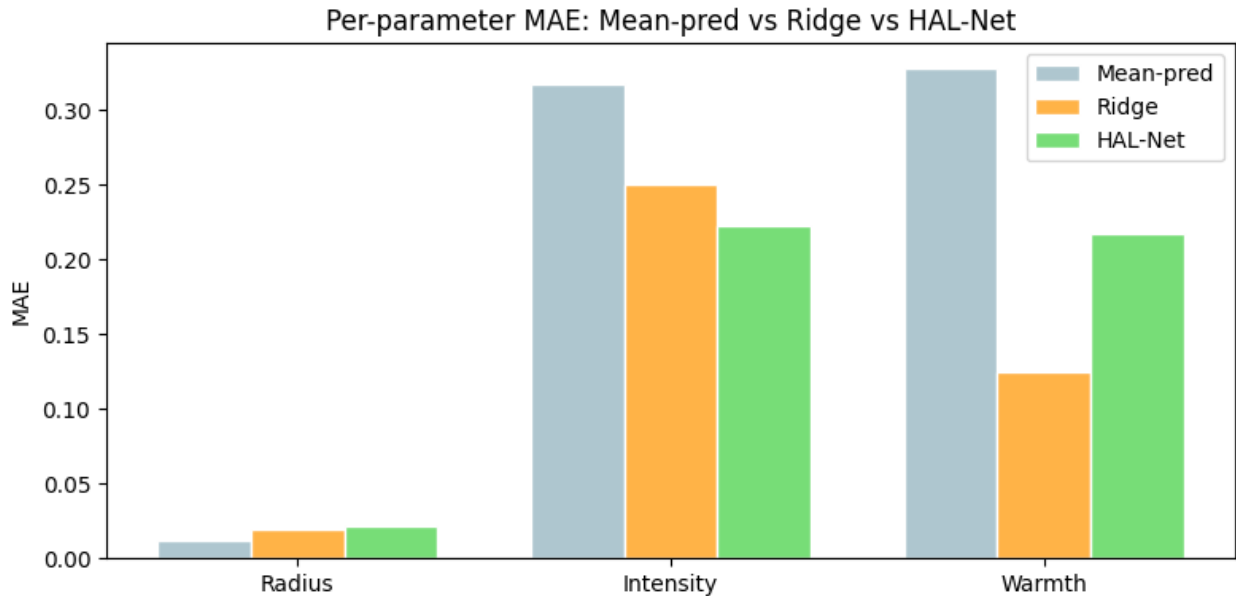


Figure 3. Per-parameter MAE: mean-prediction baseline vs ridge regression vs HAL-Net.

This result does not point in a single direction, and we report it as such rather than selecting the framing that favors HAL-Net. On intensity, HAL-Net’s 0.223 improves on ridge regression’s 0.251, the one parameter where the CNN’s added capacity translates into a measurable edge over a linear model on the same features. On warmth, the relationship reverses sharply: ridge regression reaches 0.125, against 0.218 for HAL-Net, a gap of nearly 0.1 MAE in the linear model’s favor. On radius, both learned methods underperform the mean-prediction baseline (0.019 and 0.022 against 0.012), so neither is recovering useful structure beyond what Figure 2’s near-constant radius distribution already gives away for free. Averaged across all three parameters,

ridge regression’s mean MAE of 0.132 is lower than HAL-Net’s 0.154, meaning that on this test set a 13-feature linear model is, in aggregate, a better predictor of the heuristic’s own output than the 200k-parameter CNN trained for that purpose.

The most direct reading of Table 2 is that warmth, which the original paper presented as one of HAL-Net’s two clear wins over the mean-prediction baseline, is substantially better explained by global color statistics than by anything the convolutional architecture contributes. The pseudo-labeling heuristic computes warmth from a red-versus-blue channel comparison in the halo ring (Section 3.3); a linear model given the same comparison computed over the whole image apparently captures most of the same signal, which is unsurprising once stated this way but was not tested in the original evaluation. Intensity is the parameter where HAL-Net’s nonlinear capacity earns its keep, though the margin (0.028 MAE) is modest relative to the 0.1 MAE gap by which ridge regression wins on warmth. We revisit what this implies for the network’s learned representations in Section 4.7 and for the paper’s central claim in Section 4.9.

Heuristic Constant Ablation

The most consequential constant in the pseudo-labeler is the 25% area gate: a highlight threshold level is discarded if the resulting mask covers more than 25% of the frame. Table 3 shows what happens when this gate is swept to 0.15 (stricter) and 0.40 (more permissive) on the 15 test images.

Table 3. Effect of the highlight-coverage cutoff on intensity pseudo-labels (test split, $N = 15$).

Cutoff	Images at intensity floor (≤ 0.05)
0.15	~10 / 15
0.25	~9 / 15
0.40	~7 / 15

Moving from 0.25 to 0.15 pushes one more image to the floor: a narrow real highlight gets rejected along with the diffuse content the gate is meant to catch. Moving from 0.25 to 0.40 frees three images from the floor, but at the cost of diffuse bright scenes passing through and producing inflated intensity labels. The 0.25 threshold is stable between both failure modes; the one-image difference from 0.15 is much smaller than the three-image shift at 0.40, so the choice is not arbitrary. Read alongside Section 4.4, this ablation and the floor statistics describe the same sensitivity from two angles: a meaningful share of the dataset sits right at the boundary the heuristic uses to decide whether a halo is present at all, and the network’s intensity predictions inherit whatever the heuristic decided at that boundary.

Attention Analysis

The baseline comparison in Section 4.5 raises a specific question about where HAL-Net’s intensity and warmth predictions come from. Grad-CAM (Selvaraju et al., 2017) provides a way to check this directly: it highlights the image regions whose activations most influence a chosen output, letting us see whether the network’s attention for each parameter is anchored to the halo region the heuristic measures or spread across the frame in a way more consistent with a global color statistic.

Figure 4 shows Grad-CAM maps for three test images, computed at the second residual block, for each of the three predicted parameters. The radius maps are the most spatially selective of the three: attention concentrates tightly on the highlight source itself and its immediate edge, the fire escape railing in the first row, the bright mist column in the second, and the traffic signal cluster in the third, leaving the surrounding scene

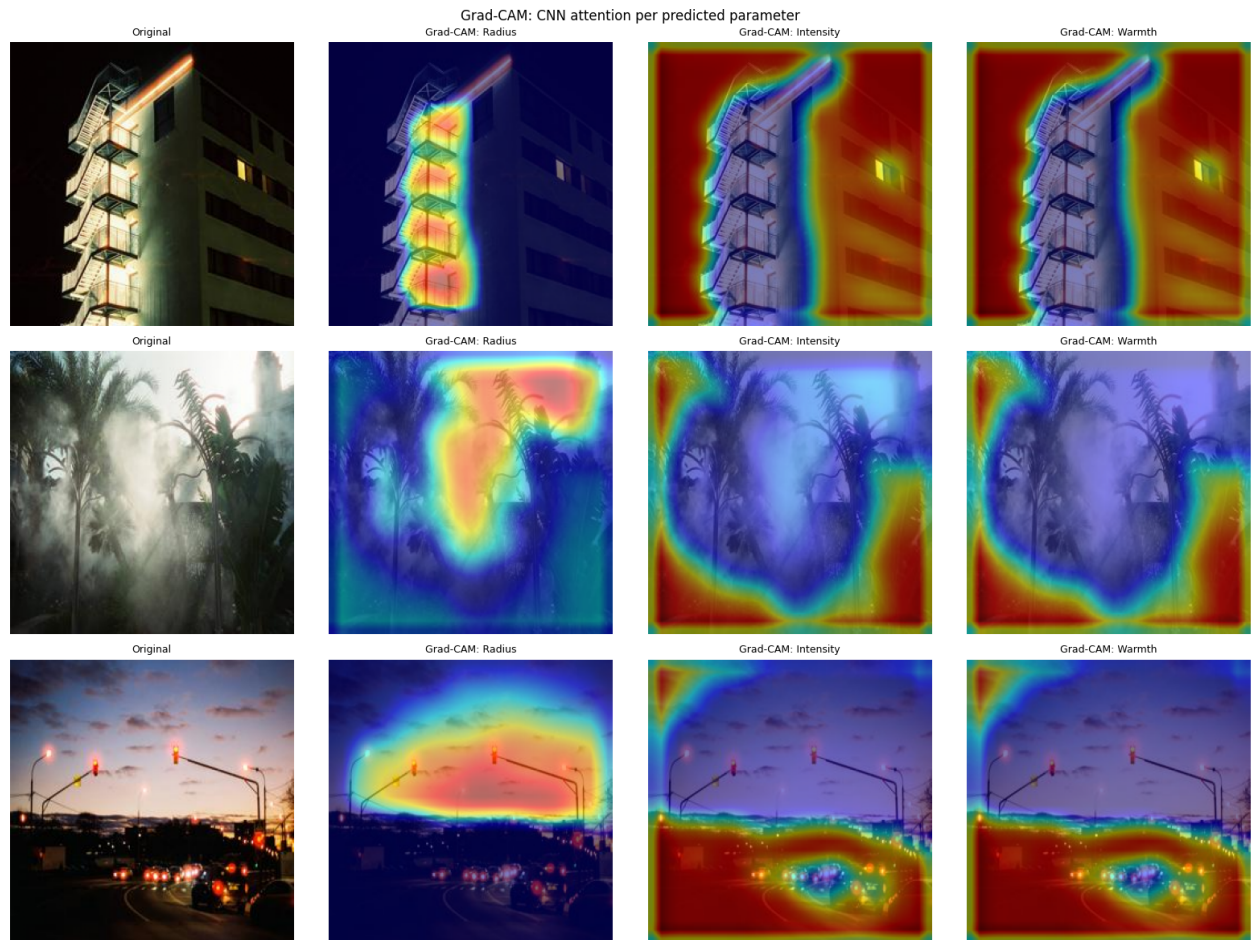


Figure 4. Grad-CAM attention for radius, intensity, and warmth on three test images.

comparatively dark in the attention map. The intensity and warmth maps look markedly different from the radius maps and largely similar to each other across all three images: both spread broadly across large regions of sky, building facade, or roadway, with comparatively little attention concentrated specifically on the halo ring around each highlight.

This pattern is consistent with, though does not by itself prove, the baseline result in Section 4.5. A network whose intensity and warmth predictions depend heavily on broad regional color and brightness, rather than on the localized halo geometry the heuristic was designed to measure, would be expected to produce diffuse Grad-CAM maps for those two parameters and tight maps for radius, which is exactly Figure 4’s pattern. It would also be expected to be closely matched, or beaten, by a linear model on global color statistics, which is what Section 4.5 finds for warmth specifically. We treat this as converging evidence from two independent diagnostics rather than as proof of mechanism; Grad-CAM reflects gradient sensitivity, not a verified causal account of what the network uses, and a full attribution study is outside the scope of this evaluation pass.

Qualitative Results

Three held-out test images were compared: the original photograph, the heuristic ground-truth halation render, and HAL-Net’s predicted halation. Images were chosen to avoid ceiling cases where intensity is pinned near its maximum.

Across all three rows in Figure 5, HAL-Net places the glow at roughly the right location and spatial extent: the fire escape railing, the traffic lights and sunset sky, and the rim-lit hair and jacket in the portrait all receive a warm halo anchored to the correct highlight. Radius estimates track ground truth closely (0.04–0.09 predicted vs 0.09 ground truth across the three examples). Warmth estimates are directionally correct but consistently lower than ground truth (0.77–0.96 predicted vs 0.99–1.00 ground truth), landing toward the orange end of the scale without fully reaching it. The clearest gap remains intensity: HAL-Net predicts 0.41–0.72 against ground-truth values of 0.50–0.75, with the largest under-prediction on the example with the strongest ground-truth glow (the traffic-light scene, 0.53 predicted vs 0.75 ground truth).

Discussion

The qualitative pattern in Section 4.8, accurate radius placement with under-predicted intensity at the high end, is consistent with the training dynamics described in Section 4.2: with 70 training images and an intensity distribution that is heavily weighted toward the floor value (Figure 2), examples near the top of the intensity range are rare, and MSE does not penalize under-prediction on rare high-intensity cases more than the same absolute error elsewhere. The model has limited incentive to commit to an aggressive intensity prediction when a conservative, average-case estimate already lowers aggregate training loss. This part of the original discussion still holds.

What the new evidence changes is the claim about warmth. The original evaluation treated warmth as one of HAL-Net’s two clear wins over a mean-prediction baseline, attributing the gain to the network learning “more stable low-level statistics (ring location and red-versus-blue channel bias) that a CNN with strided convolutions and residual blocks picks up reliably even from a small dataset.” Section 4.5 shows that a linear model given the red-versus-blue statistic directly, without any convolutional structure, beats HAL-Net on warmth by a wide margin, and Section 4.6’s Grad-CAM maps show the network’s warmth attention spread broadly across the frame rather than anchored to the halo ring the way its radius attention is. Taken together, the more accurate statement is that warmth is a parameter where the heuristic’s own underlying signal is close to linearly separable from global color statistics, and HAL-Net’s convolutional architecture is not currently extracting that signal as efficiently as a 13-feature ridge regression does. This does not mean the network has learned nothing about warmth: it improves substantially on the mean-prediction baseline (0.218 vs 0.328). It

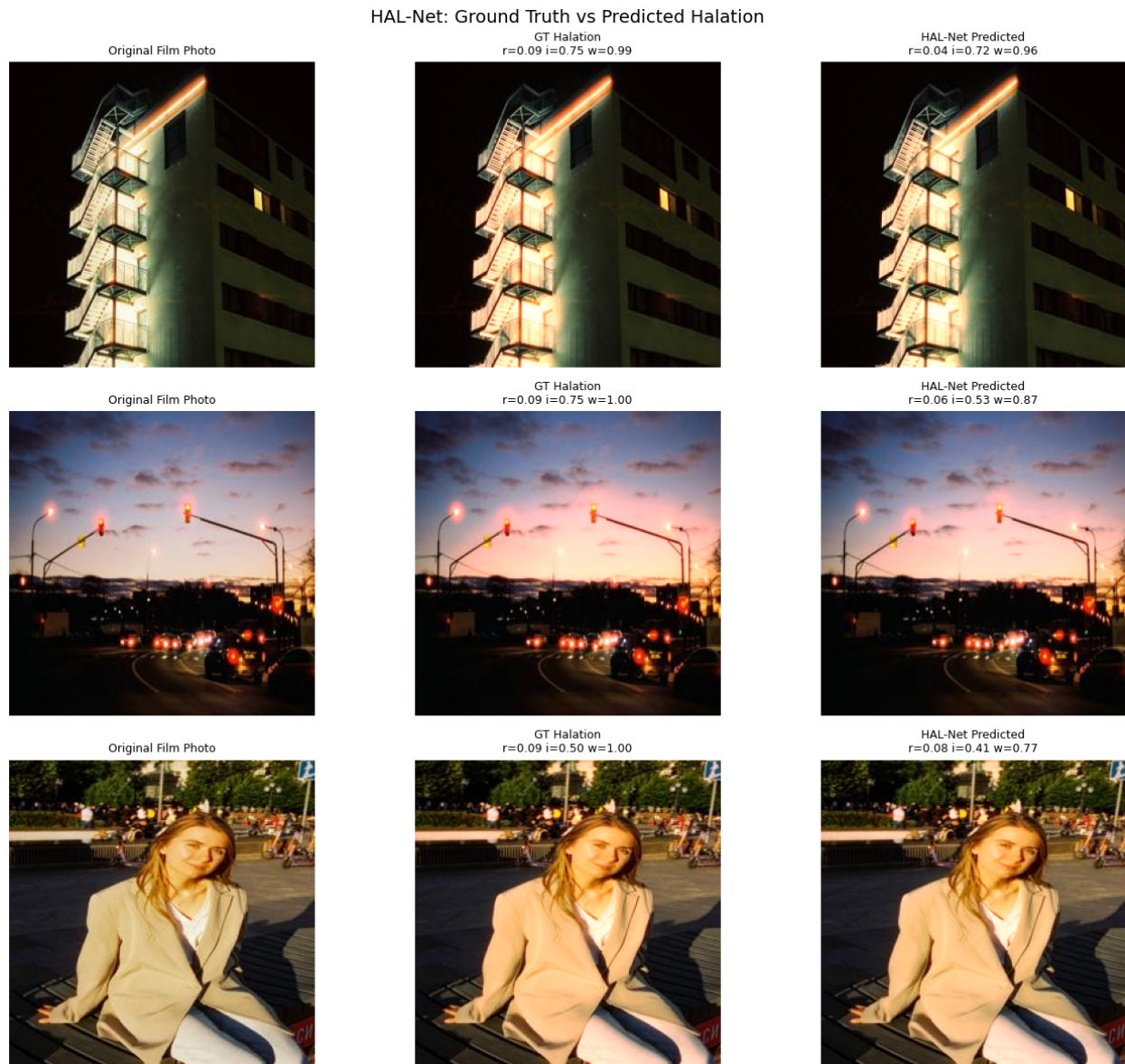


Figure 5. Ground truth versus HAL-Net predicted halation on three test images.

does mean the architecture is not yet the right tool for that specific gain, since a model an order of magnitude simpler reaches it more directly.

Intensity is the one parameter where the original claim survives intact and gains direct support: HAL-Net’s 0.223 MAE beats both the mean-prediction baseline (0.317) and the ridge regression baseline (0.251), and the Grad-CAM maps, while still broad relative to radius, are not identical to the warmth maps in every region, leaving open the possibility that some of HAL-Net’s intensity prediction draws on cues a linear model on these particular 13 features cannot represent. Radius remains the parameter neither learned method handles better than guessing the mean, for the reason established in Section 4.4: the pseudo-label distribution for radius has almost no variance to learn in the first place. Closing the intensity gap at the high end is, as before, primarily a data problem; closing the warmth gap to HAL-Net’s advantage, if that is even a goal worth pursuing given that a linear model already does better, is more likely an architecture-and-feature-design problem than a data problem, since the bottleneck Section 4.5 identifies is the network’s apparent difficulty efficiently recovering a signal a much smaller model recovers directly.

Demo and Interactive Use

HAL-Net is deployed as a Gradio application with a live instance on Hugging Face Spaces. A user uploads any image; the app pads it to square, resizes to 256×256, runs it through the trained network, and renders the predicted halation through the function in Section 3.7. The interface shows the three predicted parameters numerically alongside the rendered output. The renderer is identical to the one used for ground-truth visualization, so manual override sliders, letting a user nudge radius, intensity, or warmth away from the prediction, modify the output through the same deterministic path the model’s own predictions take. HAL-Net works as a fast initial estimate with a manual fallback, directly addressing the lack of an estimation step in the classical bloom-based tools described in Section 2.

Limitations

Ground truth is heuristic, not measured. Every label HAL-Net trains on comes from the rule-based detector in Section 3.3, not from calibrated physical measurement or human annotation. HAL-Net is best understood as a fast neural approximation to that specific heuristic; evaluating against the heuristic’s own output cannot establish how closely the heuristic (or the network trained on it) matches the physical process described in Section 1.

A linear model on handcrafted features matches or beats the CNN on two of three parameters. Section 4.5 shows that ridge regression on 13 color and brightness statistics beats HAL-Net on warmth and on mean MAE overall, and underperforms it only on intensity. This is a substantive limitation rather than a minor caveat: it means the strongest available evidence does not support a general claim that the convolutional architecture is necessary to approximate the pseudo-labeling heuristic. The Grad-CAM analysis in Section 4.6 offers a partial explanation, diffuse attention for warmth and intensity versus tight attention for radius, but does not by itself fix the gap.

Small dataset. With 70 training and 15 test images from a single film stock, the model has limited exposure to the full range of halation appearances. Section 4.9 traces the observed intensity under-prediction directly to this scarcity, and the same scarcity (Section 4.4) likely limits how informative the current ridge regression comparison can be: a 13-feature linear model needs fewer examples to fit well than a 200k-parameter CNN, so part of the gap in Section 4.5 may narrow, widen, or reverse with more training data. Generalization to other warm-toned analog stocks or to digitally simulated halation has not been tested.

No perceptual evaluation. The quantitative evaluation in Section 4 measures MAE against the pseudo-labeler’s own output. No perceptual study compares predicted and ground-truth renders, and a human preference study would be needed to establish whether the MAE differences in either direction, including the ridge regression baseline’s advantage on warmth, translate into visible quality differences.

Renderer simplicity. The halation renderer uses a single Gaussian bloom with a fixed channel-mixing rule. It does not model directional streaking, lens-dependent falloff, or multi-source interaction, all of which real film halation can exhibit around complex light arrangements.

Conclusion

HAL-Net applies the parameter-regression strategy from FGA-NN to halation, training a sub-200k-parameter CNN to predict radius, intensity, and warmth from an image’s highlight regions. Because no labeled halation dataset exists, we built pseudo-labels by distilling a hand-designed heuristic over 100 CineStill 800T scans. On the 15-image test split, HAL-Net improves over a mean-prediction baseline on intensity (0.223 vs 0.317) and warmth (0.218 vs 0.328), while radius MAE remains near the baseline due to low label variance.

Extending the evaluation with a ridge regression baseline on the heuristic’s own handcrafted features complicates this picture in a way the original evaluation did not anticipate: the linear baseline beats HAL-Net on warmth and on mean MAE, and matches it within a modest margin on the aggregate comparison, leaving intensity as the one parameter where the CNN’s added capacity shows a clear, reproducible advantage. Grad-CAM attention maps offer a consistent explanation, tight, highlight-anchored attention for radius against diffuse, frame-spanning attention for intensity and warmth, that points toward global color statistics rather than local halo geometry as the dominant signal behind the network’s warmth predictions specifically.

We see this as a clarification of HAL-Net’s contribution rather than a reversal of it. The network reliably localizes the halo (radius) and provides a real, if partial, improvement on intensity over both baselines; it does not currently provide evidence of outperforming a far simpler model on warmth, and any future claim about the architecture should be scoped to the parameter where the evidence supports it. The clearest next steps are expanding the training set at the high-intensity end, where Section 4.9 traces the largest remaining gap, and either redesigning the warmth pathway to make better use of the convolutional features it already computes or, if a linear model continues to win, replacing the warmth head with one.

References

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